BBA - Proteins and Proteomics 1869 (2021) 140659

Contents lists available at ScienceDirect

BBA - Proteins and Proteomics

journal homepage: www.elsevier.com/locate/bbapap

Research Paper

Meta-omics analysis indicates the saliva microbiome and its proteins associated with the prognosis of oral cancer patients☆

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ARTICLE INFO

Keywords:
Saliva
Mouth neoplasms
Oral microbiome
Mass spectrometry
Metagenomics
Diagnosis
Prognosis

ABSTRACT

Saliva is a biofluid that maintains the health of oral tissues and the homeostasis of oral microbiota. Studies have demonstrated that Oral squamous cell carcinoma (OSCC) patients have different salivary microbiota than healthy individuals. However, the relationship between these microbial differences and clinicopathological outcomes is still far from conclusive. Herein, we investigate the capability of using metagenomic and metaproteomic saliva profiles to distinguish between Control (C), OSCC without active lesion (L0), and OSCC with active lesion (L1) patients. The results show that there are significantly distinct taxonomies and functional changes in L1 patients compared to C and L0 patients, suggesting compositional modulation of the oral microbiome, as the relative abundances of Centipeda, *Veillonella*, and *Gemella* suggested by metagenomics are correlated with tumor size, clinical stage, and active lesion. Metagenomics results also demonstrated that poor overall patient survival is associated with a higher relative abundance of *Stenophotromonas*, *Staphylococcus*, *Centipeda*, *Selenomonas*, *Alloscordovia*, and *Acitenobacter*. Finally, compositional and functional differences in the saliva content by metaproteomics analysis can distinguish healthy individuals from OSCC patients. In summary, our study suggests that oral microbiota and their protein abundance have potential diagnosis and prognosis value for oral cancer patients. Further studies are necessary to understand the role of uniquely detected metaproteins in the microbiota of healthy and OSCC patients as well as the crosstalk between saliva host proteins and the oral microbiome present in OSCC.

1. Introduction

Oral cancer is a major cause of morbidity and mortality among cancer types, with an estimated incidence of more than 300,000 cases and 145,000 deaths worldwide in 2018 [1]. The most frequent type is the oral squamous cell carcinoma (OSCC), representing over 90% of oral malignancies [2,3]. OSCCs are of epithelial origin and initially arise in

☆ This article is part of a Special Issue entitled: The "Wave" of Brazilian Proteomics edited by Dr. Daniel Martins-de-Souza.

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https://doi.org/10.1016/j.bbapap.2021.140659
Received 1 December 2020; Received in revised form 4 April 2021; Accepted 5 April 2021
Available online 9 April 2021
1570-9639/© 2021 Published by Elsevier B.V.

the oral cavity or oropharynx as localized and usually well delimited erythroleukoplastic lesions [4,5]. Due to its fast progression, the early diagnosis of OSCC has a critical role in its prognosis, directly affecting treatment effectiveness and the survival rates.

Abbreviations

OSCC	Oral Squamous Cell Carcinoma
N0	Absence of lymph node metastasis
N+	Presence of lymph node metastasis
ROC-AUC	Area Under the Receiver Operating Characteristic curve
TNM	Tumor (T), Nodes (N), Metastasis (M) classification system
LC-MS/MS	Liquid Chromatography-Tandem Mass Spectrometry
FDR	False Discovery Rate

Well-established etiological agents and risk factors for OSCC include tobacco usage, betel chewing, alcohol consumption, and human papillomavirus (HPV) infection (in the specific context of oropharyngeal carcinomas), in which the latter presents different clinical and treatment behavior [3,5]. When combined with tobacco or alcohol use, poor oral hygiene acts synergistically to increase the risk factors influence and lead to chronic infection and inflammation, both of which are recognized cancer hallmarks and can act as important drivers of the tumor development [6].

The importance of the human microbiome within the body has been more intensively addressed in the past years. In health, the microbiome interacts with the host and contributes to normal tissue functioning as well as to the protection from pathogen colonization, by maintaining the balance of commensal bacteria [7,8]. However, if that commensal community is disrupted, a dysbiosis occurs and can contribute to the pathogenesis of different types of carcinomas [9]. Bacteria have been linked to cancer through several mechanisms such as chronic inflammation, antigen-driven lymphoproliferation, induction of hormones that increase epithelial cell proliferation, and production of toxic or carcinogenic metabolites [10,11]. Alteration in the composition, diversity and function of the oral microbiome has been described in oral cavity diseases, as periodontitis and dental caries, in which microbiome dysbiosis is associated with poorly regulated immune responses that lead to inflammatory processes [8,12]. Periodontitis is an example of an inflammatory event where polymicrobial synergy and dysbiosis components play a role in its development, and where the pro-inflammatory T helper cells (Th17) have been highlighted as critical regulators of the barrier immunity [12–14].

Several studies have found a small but positive relationship in the association between periodontitis outcomes (e.g, gum bleeding, teeth loss) and oral cancer [15–18]. The underlaying mechanisms are related to the formation of carcinogenic metabolites, such as the transformation of nitrate into nitrite or the production of acetaldehyde by bacterial species present in the oral microbiome of patients with periodontitis [19]. Other possible pathways by which oral bacteria may contribute to oral carcinogenesis include inhibition of apoptosis, activation of cell proliferation, promotion of cellular invasion, induction of chronic inflammation and production of carcinogens [20,21]. Previous studies found that *Porphyromonas gingivalis* leads to anti-apoptotic processes in human cells [22,23] and activates gingival epithelial cell progression by accelerating the cell cycle [24,25], both cancer hallmarks events. Furthermore, *P. gingivalis* is involved in OSCC invasion by the expression and activation of proMMP-9 [26,27]. In a similar way, *Fusobacterium nucleatum* infected cells may activate multiple cell signaling systems that promotes an increase in the expression of collagenase 3 (MMP-13) [28]. Inflammatory factors, (e.g chemokines, cytokines, reactive oxygen and nitrogen radicals) are believed to contribute to carcinogenesis by inducing cell proliferation and promoting tumor cell inhibition of apoptosis through activation of oncogenes and inactivation of tumor-suppressor genes [29]. *P. gingivalis* presented pro-inflammatory properties, upregulating the receptors B7-H1 and B7-DC, known to contribute to chronic inflammation [30]. Other oral bacterial species, such as *Streptococcus salivarius*, *Streptococcus intermedius*, *Streptococcus mitis*, and *Candida spp* presented significant alcohol dehydrogenase-enzyme activity, playing a role in metabolizing ethanol to carcinogenic acetaldehyde in the mouth under aerobic or microaerophilic conditions [31,32].

Despite these discoveries, few studies explored the oral microbiome in different stages of oral carcinogenesis to determine not only a difference between healthy and cancer patients, but if the presence of specific microorganisms is related to clinicopathological features relevant to early diagnosis and prognosis determination. Previous studies from our group have shown that the analysis of saliva proteome can reveal immune processes related to the presence of OSCC and indicate that proteomics can contribute to determining OSCC prognosis and treatment outcomes [33,34]. However, there are no clear evidence that the microbiome composition in saliva can be modulated or modulates OSCC and whether its composition associates to survival outcome. Therefore, screening the microbiome composition of OSCC saliva samples by a combination of metagenomics (MG) and metaproteomics (MP) approaches may be an interesting strategy in the quest for understanding the microbiome and its role in oral cancer onset and progression. Also, unveiling the role of oral microbiome to OSCC clinical outcomes, as well as the evolving interactions between microbiota, immune system and tumor, may help elucidating OSCC cases in which traditionally recognized well established risk factors are not present [35].

In this study, we examined the microbiome taxa composition and the host-microbiome proteome repertoire of 16 oral cancer patients and 8 control cases to compare and explore the microbial role in OSCC and infer clinicopathological microbiome and protein profiles. Using 16S rRNA miSeq and mass spectrometry quantitative techniques, we determined the microbiome profiles and the proteome of control individuals as well as of OSCC patients with and without active oral tumors (L1 and L0, respectively), and investigated the association of genus levels and protein abundance with patient clinicopathological data. Our results indicated alterations in the oral microbiome composition and abundance in all patient groups and their correlation with clinical features. Besides, the microbiota proteins, were able to contribute with complementary features regarding the microbial composition and its functional characteristics. Altogether, these data indicate that both metagenomics and metaproteomics analysis of saliva have a good potential to assist the diagnosis and prediction of OSCC treatment outcome.

2. Material and methods

2.1. Patient selection

This study comprised a two patient's cohort, including control and OSCC patients. The detailed clinical and pathological information for the 16 OSCC patients enrolled in this study are summarized in Supplementary Table 1. This study was approved by the Ethics Review Board of the Instituto do Câncer do Estado de São Paulo (ICESP), São Paulo, SP, Brazil, and Plataforma Brasil through protocol CAAE 30658014.1.1001.0065 and by the Ethics Review Board of the Faculty of Odontology of Piracicaba (FOP), São Paulo, SP, Brazil, and Plataforma Brasil through protocol CAAE 51762415.5.0000.5418. All participants provided a written informed consent. The study was conducted following the declaration of Helsinki and was performed following the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement [36]. The procedures used for saliva sampling [33] and annotation were performed following the guidelines and

experimental protocols approved by both ethics committees.

2.2. Saliva sampling

Saliva samples were obtained from healthy control individuals (n = 8) and OSCC patients (n = 16), including patients who had undergone surgical resection (designated as "without active lesion" or L0, n = 8) and patients who had active oral malignant lesions (designated as "active lesion" or L1, n = 8) at the time of saliva collection. In average the period between resection and saliva collection was around 30 days. Individuals were advised not to eat for at least 1 h, then rinsed their mouths with 5 mL of drinking water, and saliva was subsequently harvested without stimulation into a plastic sterile bottle [33]. The saliva samples were aliquoted into 15 mL tubes and frozen at −80 °C for long-term storage.

2.3. Sample preparation for metagenomics

2.3.1. 16S rDNA extraction from saliva samples

About 5 μL of saliva from each of the 24 patients was used to prepare triplicate libraries for taxonomic and diversity analyses. The hypervariable V4 region of the 16S rRNA was amplified by polymerase chain reactions (PCR) using the universal primers 515F (5'-GTGCCAGCMGCCGCGGTAA) and 806R (GGACTACHVGGGTWTCTAAT) attached with specific adapter sequences for multiplexing during sequencing. The amplicons were purified using Agencourt AMPure XP purification system kit (Beckman Coulter, USA) and the samples were indexed with Illumina Nextera indexes kit (Illumina INC.). The libraries were quantified by qPCR using the KAPA Library Quantification Kits for Illumina sequencing platforms and, sequenced on Illumina Miseq using a 2 × 300 bp paired-end sequencing kit, available at LNBR/CNPEM NGS facility.

2.3.2. Sequencing analysis

Raw reads were quality assessed using FASTQC [37] (https://www.bioinformatics.babraham.ac.uk/projects/fastqc/), and preprocessed by removing adapters and low-quality sequences, using Trimmomatic [38]. Quality-filtered reads were merged using fastq_mergepairs function of Usearch v.8 package [39] and further analyzed using the UPARSE pipeline [40]. The sequences were clustered into operational taxonomic units (OTU) based on a 97% identity using the USEARCH cluster_otus function (Supplementary Table 2). Chimeric sequences were identified and removed using the UCHIME method. The representative sequence from each OTU was assigned to taxonomic taxa using Ribosomal Database Project and RDP Classifier [41] (Supplementary Table 3). For statistical analyses, the Chao1, ACE, Shannon, and Simpson indexes, rarefaction curves, and sample coverage (Good's coverage) were calculated by employing the Phyloseq and Vegan packages on the R statistical software version 3.2. For genera relative quantification, the contribution of OTUs from the same genus were classified regarding the total number of reads and the normalized data was used for statistical analysis (Supplementary Table 4). To perform relative quantification in OTU level, the normalized OTU data was used (Supplementary Table 5). Attribute selection was performed using the OTU table to define 12 genera out of 19 OTUs that better depict the sample groups (Supplementary Tables 6–7). Additionally, data classification and feature selection analysis were performed using the Weka v.3 suite (http://www.cs.waikato.ac.nz).

2.4. Sample preparation for host proteomics and Metaproteomics

2.4.1. Cell disruption and isolation of supernatant proteins from saliva samples

Saliva preparation was performed using slightly modified version of a previously published protocol [34]. Briefly, saliva was initially centrifuged for 5 min at 1500 ×*g*, 4 °C to separate intact cells and debris, (designated as cells), from the supernatant fraction. The supernatant, containing the soluble or secreted proteins, was separated in a new tube for protein concentration determination. The proteins were quantified using a Bradford assay kit and the volume corresponding to 10 μg was added to 8 M urea.

The saliva cells were resuspended and homogenized in 200 μL of urea buffer (100 mM Tris-HCl pH 7.5, 8 M urea, and 2 M thiourea) containing the cOmplete Mini Protease Inhibitor Cocktail, 5 mM EDTA, 1 mM PMSF and 1 mM dithiothreitol (DTT). An ultrasound treatment was carried out for 5 × 30 s on ice (with 50% of the device power enabled) to disrupt the cells in the cellular fraction followed by another centrifugation step (5 min, 4 °C, 10,000× *g*). The supernatant was stored on ice for further analysis. The proteins extracted from the saliva cells were quantified using a Bradford assay kit, and the volume corresponding to 10 μg was used for further trypsin digestion.

2.4.2. MS sample preparation (supernatant and cellular fractions)

For protein digestion, proteins extracted from supernatant and cell fractions were treated with dithiothreitol (DTT 5 mM for 25 min at 56 °C) and alkylation was performed with iodoacetamide (IAA 14 mM for 30 min at room temperature in the dark). For protein digestion, urea was diluted to a final concentration of 1.6 M with 50 mM ammonium bicarbonate, and 1 mM of calcium chloride was added to the samples for trypsin digestion for 16 h at 37 °C (0.2 μg of trypsin for 10 μg of total protein). The reaction was quenched with 0.4% formic acid, and peptides were desalted with C18 stage tips, dried in a vacuum concentrator, reconstituted in 0.1% formic acid and stored at −20 °C for subsequent analysis by LC-MS/MS.

2.5. LC-MS/MS analysis

Proteolytic digestion was performed with trypsin on 48 samples and analysis was carried on using nano-LC-MS/MS (Supplementary Table 8). One microgram of the peptide mixture in 4.5 μL was analyzed using an LTQ Orbitrap Velos (Thermo Fisher Scientific, San Jose, CA) mass spectrometer coupled to nanoflow liquid chromatography on an EASY-nLC system (Proxeon Biosystems, Odense, DK) with a Proxeon nanoelectrospray ion source. Peptides were subsequently separated in a 2–90% acetonitrile gradient in 0.1% formic acid using a PicoFrit analytical column (20 cm × ID75, 5 μm particle size, New Objective) at a flow rate of 300 nL/min over 80 min, in which a gradient of 35% acetonitrile is reached in 45 min. The nanoelectrospray voltage was set to 2.2 kV, and the source temperature was set to 275 °C. The instrument methods employed for LTQ Orbitrap Velos were set up in data-dependent acquisition (DDA) mode. Full scan MS spectra (m/z 300–1600) were acquired in the Orbitrap analyzer after accumulation to a target value of $1e^6$. Resolution in the Orbitrap was set to $r = 60{,}000$, and the 20 most intense peptide ions (top 20) with charge states ≥2 were sequentially isolated to a target value of 5000 and fragmented in the high-pressure linear ion trap by CID (collision-induced dissociation) with a normalized collision energy of 35%. Dynamic exclusion was enabled with an exclusion list size of 500 peptides, an exclusion duration of 60 s and a repetition count of 1. An activation Q of 0.25 and an activation time of 10 ms were used. A complete overview of the experimental workflow is given in Fig. 1.

2.6. Proteomics data analysis

To evaluate our metaproteomics data, we used a two-step data analysis: i) First, we built two different databases retrieved from human oral microbiome database (HOMD) [44] and Uniprot sources. For the first database, we obtained all full-length fasta protein sequences of all named species that belonged to the 12 feature selected genus retrieved in the 16S rDNA sequencing (downloaded 11 June 2019; 1,163,032 bacterial protein sequences in total) merged with the human Uniprot protein database (93,599 protein sequences, 36,574,184 residues,

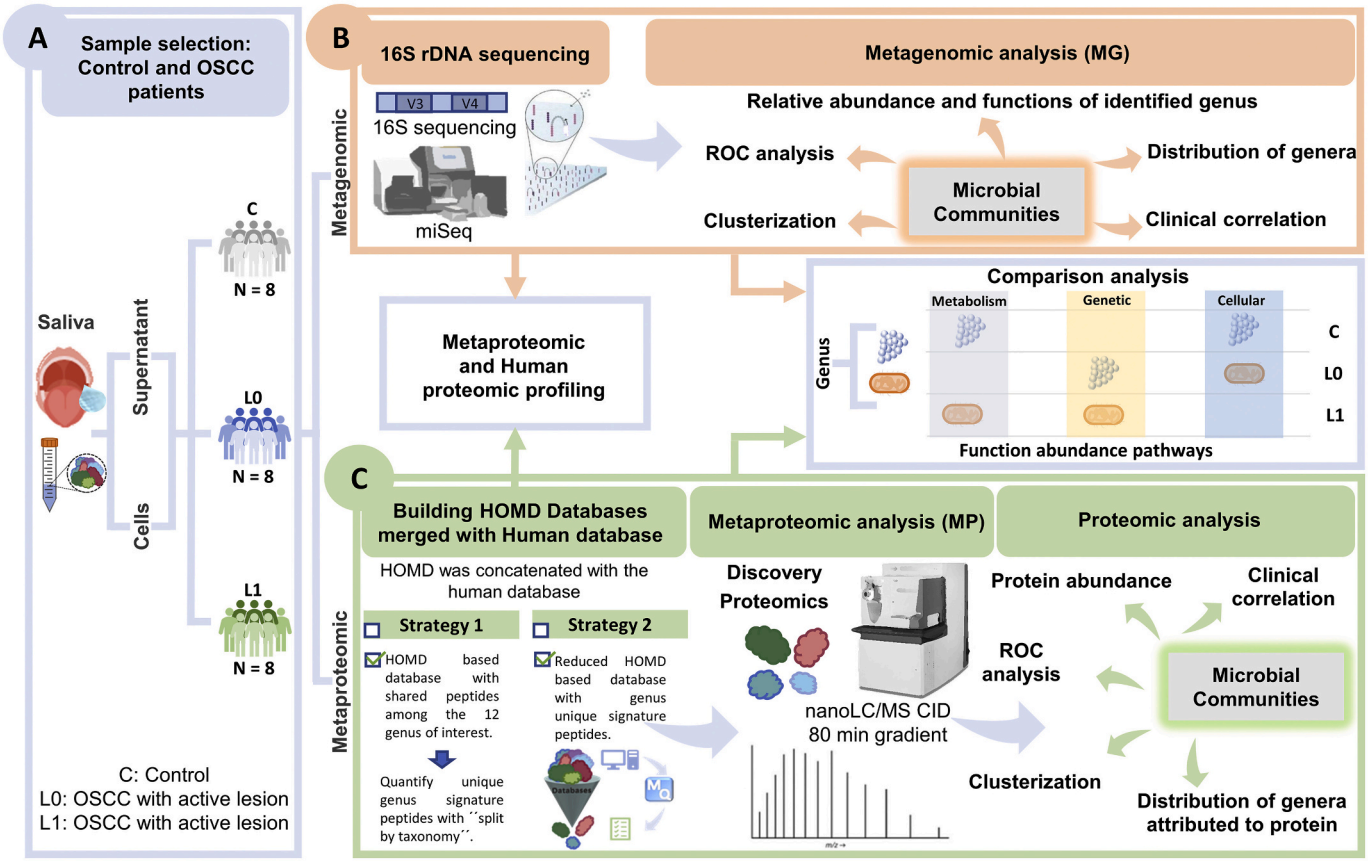


Fig. 1. Metagenomics and metaproteomics strategies of saliva indicate that microbiota and its proteins correlate with prognostic factors of oral cancer patients. Saliva samples were analyzed by microbial and protein profiling. The combination of both analyses enabled the identification of significant differences in the abundance of microbial taxa and functions and alterations in the abundance of human proteins. a) This study was nested within a cross-sectional cohort study of 24 individuals, of whom 16 were OSCC patients and the saliva was collected before the surgery (L1) or after surgery (L0), besides, 8 healthy individuals were included as controls (C). Saliva cells were separated from supernatant proteins by centrifugation and both cells and supernatants were analyzed. b) Microbiome profiling was performed in total microbial DNA of saliva cells. The hypervariable V4 region of the 16S rRNA and internal transcribed spacer region 2 (ITS2) were amplified using universal primers [42,43]. PCR products were sequenced in an Illumina MiSeq system. Based on the identified genus, statistical analysis, hierarchical clustering analysis, and correlation analysis with clinical features were performed. In addition, the genera were tested as potential classifiers of Control and OSCC patients using ROC curves. c) Based on the genus determination given by metagenomics analysis, two protein databases were generated to search the LC-MS/MS proteomic data derived from the cells and supernatant proteins in saliva. A total of 187 meta-proteins and 569 human proteins were identified in this study. One database contains all sequences from the Human Oral Microbiome Database (HOMD) that matched with the genus identified in the 16S rDNA sequencing and selected by feature attribution and the other database contains the same genus but we have considered only the unique tryptic sequences from each genus to generate the database. Both databases were merged with the Human protein database and used to search against the proteomic data. Based on the host and meta-proteomics data, statistical analysis, hierarchical clustering, and correlation analysis with clinical features were performed. The differential host and bacterial proteins were tested as classifiers of Control and OSCC patients using ROC curves. Comparison analysis considering both MG and MP data was performed at the genus and functional level.

downloaded in June 2018) and uploaded in the Andromeda search engine [45]. For the second database, the HOMD was processed basically as described in [46], with the purpose of reducing the presence of redundant peptides among different genus. For that, we performed an in silico trypsin digestion (KR, not P) of HOMD protein sequences that composed the first database, filtered out peptides smaller than 7 residues long, then removed tryptic peptides shared by two or more bacteria genus. This list of genera-specific tryptic peptides was concatenated to reconstruct the protein sequences then merged with the human Uniprot database (1,160,275 protein sequences in total); ii) Secondly, meta-proteomics data was submitted to MaxQuant version 1.5.3.12 with the 48 LC-MS/MS runs corresponding to the supernatant and cellular fractions. Target-decoy database search was performed at a maximum of 1% false discovery rate (FDR) for both protein and peptide identification and label-free quantification was set using the MaxLFQ algorithm [47]. Both razor and unique peptides were used for protein quantification, and the minimum ratio count was set as 1. A match time window of 2 min was applied. As search parameters, a mass tolerance of 6 ppm was considered for precursor ions and 0.5 Da for fragment ions, enzyme specificity was set to trypsin/P, with a maximum of two missed cleavages for the first database and none for the second database search. Carbamidomethylation of cysteine was considered as fixed modification, and oxidation of methionine and protein N-terminal acetylation were considered variable modifications. Only for the first built database has the "split by taxonomy" function been enabled [47].

Statistical analysis was performed with Perseus v.1.6.5 software [48], which is available in the MaxQuant package. Identified protein entries from saliva cell (Supplementary Tables 9–10) and supernatant (Supplementary Tables 11–12) samples were processed, excluding reverse sequences and those entries identified "only by site". Contaminants were not removed from the dataset because keratin proteins are of interest in the study. Protein abundance, which was calculated based on the normalized spectrum intensity (LFQ intensity), was log2-transformed, and the dataset was filtered by one minimum valid value in at least one group for HOMD shared peptide and non-redundant genus peptide databases, respectively. Significance was assessed using Student's *t*-test or One-way ANOVA (P-value < 0.05) (Supplementary Tables 9–12). Exclusive and common proteins from each comparison (C, L0

and L1) and between each database are presented as a Venn diagram generated using the InteractiVenn tool [49]. For data visualization, heat maps with *z*-score values of log2 LFQ intensities and volcano plots were built using the open-source R statistical programming language.

2.7. Association of clusters, genus and OTUs relative abundance and host-bacterial protein abundances with clinicopathological patient data

Linear regression analysis was performed using R to evaluate the linear relationship between genus relative abundance and protein abundances with clinicopathological features namely gender, lesion type, tumor site, TNM clinical and pathological (cT/pT, cT/pT grouped, cN, cN grouped), and age. The presence of periodontitis was not assessed in this study. Also, linear regression analysis was used to assess the linear relationship between cluster and genus abundance. Linear regression with P-value < 0.05 after Mann-Whitney or Kruskal-Wallis was used to define significance. The Pearson product-moment correlation coefficient (denoted by R) was also calculated to measure the strength of the association (i) between clusters and clinicopathological features, (ii) between genus/OTUs/LFQ and clinicopathological features, (iii) between cluster and genus relative abundance. Only genus and proteins with $-0.38 > R > 0.38$, and P-value < 0.05 were considered for the final analysis (Supplementary Tables 13).

Correlations between the clusters and the clinicopathological data were evaluated by the Pearson product−moment correlation coefficient and cross-tabulation with the chi-square test (Supplementary Table 13). P-value < 0.05 was used to define significance.

A survival analysis was also performed using Kaplan−Meier methodology and compared with the log-rank test (Supplementary Table 14). In the univariate survival analysis, the comparison was performed between the higher and the lower expression values. These were defined by group formation considering log2 expression of the proteins of interest and the genus relative abundance present in the 16 OSCC patients with the information of patients' overall survival status.

2.8. ROC curve by logistic regression and random forest analysis

The power of each genus and protein LFQ abundance as classifiers of control individuals and OSCC patients was evaluated by the construction of ROC curves (Receiver Operating Characteristic) using Random Forest and Logistic Regression. The sensitivity and specificity were calculated using the genus relative abundances or protein intensities from label-free quantification experiments. The area under the curve (AUC-ROC) with a 95% confidence interval was used for comparison (Supplementary Table 15). Optimal cut-off by highest sensitivity (True Positive Rate) as function of the specificity (False Positive Rate) was calculated, and the decision threshold was assigned to the value of 70%. For all statistical comparisons, an ANOVA P-value ≤ 0.05 was considered as significance threshold. The data analysis was performed using the package *pROC* [50] and the R environment version 3.4.4. Also, the combination of the MG data and proteomics data was performed to evaluate the ability to increase the group separation. For that, we considered the relative quantification of the four genus taxonomies (*Granulicatella*, *Selenomonas*, *Veillonella*, and *Rothia*) that showed the best AUC ROC values and combined with the four host proteins (S100A-9, ELNE, CTSG, BPIB1) with AUC ROC $> 70\%$. The raw data was converted to a range from 0 to 1 for comparison, and the AUC ROC curves were performed using the packages scipy [51] and scikit-learn [52]. The value 0.3 was used as a train-test split during calculations.

2.9. Statistical analysis and functional characterization

Control, L0 and L1 patients were grouped according to molecular profiles using *Nonmetric multidimensional scaling* analysis *(NMDS)* and partial least squares discriminant analysis (PLS-DA) in the web-server Metaboanalyst 4.0 [53], and hierarchical cluster heat maps in R environment. The distribution and identification of changes in proteomics dataset were displayed in volcano plots also built in R language. A P-value ≤ 0.05 was used to determine significance in functional characterization analysis. Functional prediction from 16S rDNA sequencing data was performed by PICRUSt2 analysis [54]. We have used the Burrito web software [54] to visualize the taxonomic and functional compositions of multiple samples and to highlight the dynamic relationships between taxa and functions to capture the underlying structure of these data (Supplementary Table 16). In addition, to display the enriched host biological processes and molecular functions by Gene Ontology terms and the interaction among the differential human proteins we have used EnrichR [55] and STRING [56], respectively.

3. Results

3.1. Experimental design combining meta-omics

Starting from the premise that saliva plays a role in regulating oral microbiota, we analyzed the effect of OSCC on the microbial communities in the oral cavity through a characterization of the local microbiome. Following the experimental design described in Fig. 1, we investigated whether oral cancer and its changes to the host proteome were able to alter the microbial communities in the oral cavity.

First, we performed a characterization of the local microbiome. We collected saliva samples, as described in the section on materials and methods, and the oral microbiome was characterized using saliva cells containing host and microorganism cells.

We performed 16S rDNA gene sequencing with the saliva-derived samples. We acquired a set of 18 million raw reads sequencing the 16S rDNA V4 hypervariable regions of the 24 DNA samples used in the study. The number of sequences per sample ranged from 130,000 to 345,000 (268,000 sequences on average) (Supplementary Table 2). The sequences underwent strict quality and size filtering to remove reads shorter than 150 bp as well as those with mismatches and low-quality scores. Sequences were error-corrected using the Usearch tool, and this was followed by de novo chimera detection with the Uchime program and screening for contaminant sequences. In total, around six million sequences passed preprocessing. These sequences were clustered into 617 OTUs (Supplementary Table 3). The abundance was normalized to the total amount of OTUs in each sample. Sample S99 (L0) proved to be an outlier in these data and was excluded from further metagenomic analysis. Additionally, low-abundance OTUs with less than one read in at least 5% of samples were removed, resulting in 284 OTUs (Supplementary Table 5).

3.2. Oral cancer (with and without active lesions) alters the relative abundance and taxonomic functional profiles of different microorganisms in saliva

To compare the composition of the microbial community in the oral activities of different patients, we took into account the specific bacterial genera present as well as their relative abundances within the sample groups being compared. We first determined the metagenomic profiles of the three different sample groups (C, L0, and L1), considering all 72 genera identified in the 284 OTUs that passed the quality control cutoff (Fig. 2A). *Streptococcus*, *Veillonella*, *Prevotella*, *Neisseria,* and *Fusobacterium* are among the most abundant microorganisms in saliva, assigned at the genus taxonomy level (Fig. 2B). We further applied Student's *t*-test to detect statistically significant differences between the relative abundances of the genera identified: C vs S (L0 + L1), C vs L0, C vs L1, and L0 vs L1 (Table 1, Student's t-test, $p < 0.05$). The relative abundance of all 72 genera did not cluster according to the three biological groups, but 15 taxonomical groups (genera) with differences in abundance (Student's t-test, P-value < 0.05, C vs S, C vs L0, C vs L1, and L0 vs L1, Table 1) could be clustered among the control and patient samples—one cluster being mainly formed by control samples and the other two clusters by oral

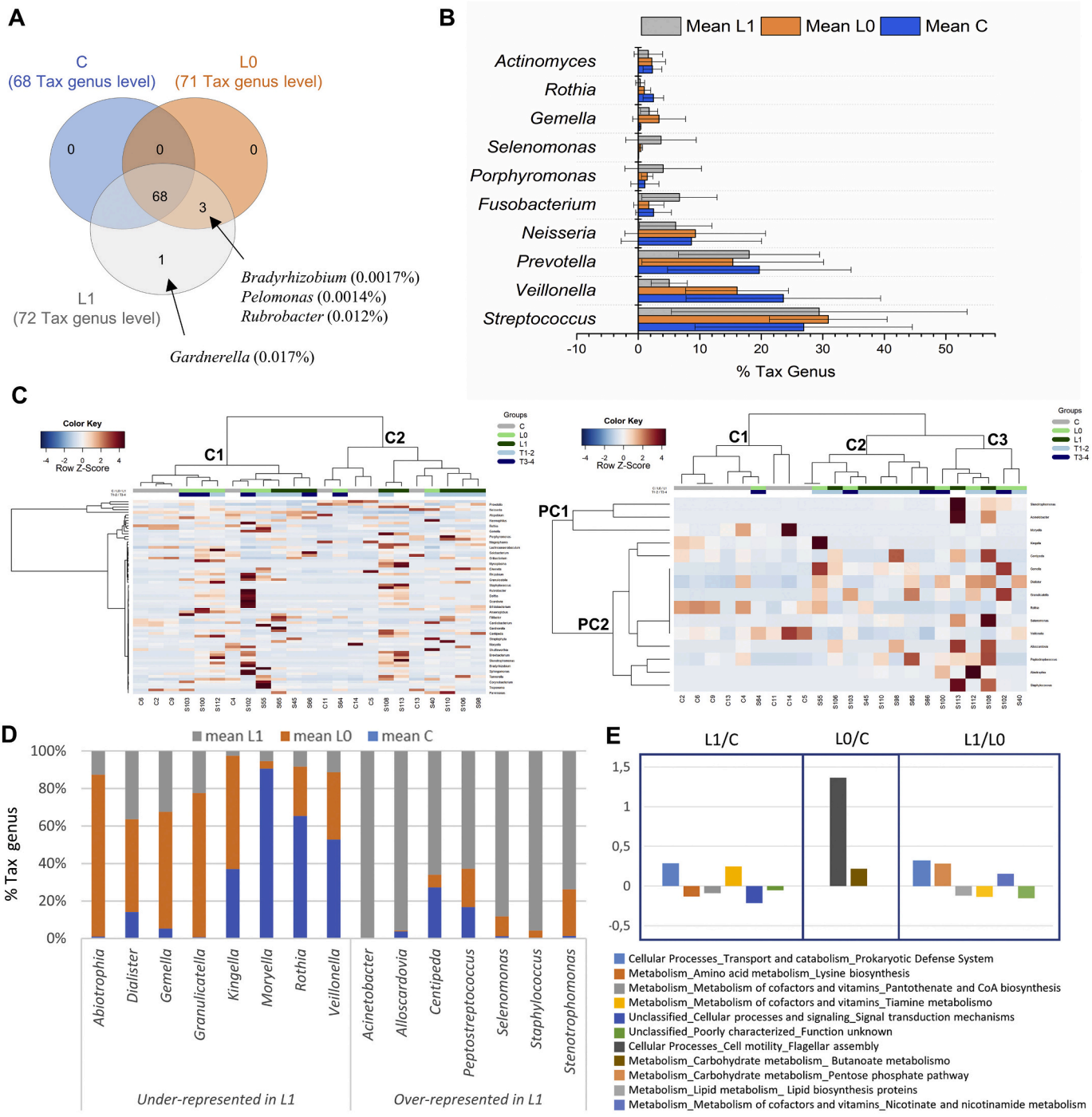


Fig. 2. Differential taxonomy distribution at genus level between control individuals, L0 and L1 patients. (A) Distribution of OTUs classified at genus level (72 different taxonomies at genus levels) with high overlap between Control individuals, L0 and L1 patients. Only one genus was found exclusive to L1 and three were found exclusive to OSCC (L0 and L1), but all four of them presented low contribution to the total microbiome, less than 0.1%. (B) Genera distribution profile (%) in control individuals, L0 and L1 patients of the 10 most abundant genus identified in cellular saliva. *Streptococcus*, followed by *Veillonella*, *Prevotella* and *Neisseria* are among the most abundant genera in Control individuals and L0 patients, while in L1 patients *Streptococcus*, followed by *Prevotella*, *Fusobacterium* and *Neisseria* are among the most abundant genera. (C) Hierarchical clustering considering all 72 genera (left side) indicates the formation of two main clusters, without separating Control and OSCC, or OSCC patients. Among the 72 identified genera, 15 of them are differential and below the significance threshold (Student's *t*-test, $p < 0.05$) comparing control individuals, L0 and L1 patients. When considering only the 15 differential genera (right side) the clustering separates better OSCC patients from control individuals. Both hierarchical clusters were performed in R environment with the Euclidean Ward D. (D) The differential genera can be divided in enriched in control and L0 compared to L1 and enriched in L1 patients compared to control and L0. (E) Differential sub-pathways annotated in Burrito are shown comparing Control-L0, Control-L1 and L0-L1 (Student's *t*-test, $p < 0.05$). Color bars show log2FC values of the 11 differential sub-pathways inferred by the metagenomic data.

Table 1
Differential taxonomies inferred by 16S rDNA abundance in control, L0 and L1 saliva at the genus level (%) (C vs S; C vs L0; C vs L1; L0 vs L1; Student's t-test, p < 0.05).

Microorganism (genus taxonomy)	Comparison	Student's *t*-test P-value	Log2 FC
Abiotrophia	L1 vs Control	0.01	3.56
Acinetobacter	L0 vs Control	0.04	3.60
Alloscardovia	L1 vs Control	0.03	4.64
	L1 vs L0	0.04	7.79
Centipeda	L0 vs Control	0.04	−1.98
Dialister	S vs Control	0.04	1.58
	L0 vs Control	0.04	1.80
Gemella	L1 vs Control	0.01	2.60
Granulicatella	L0 vs Control	0.008	6.86
Kingella	L1 vs Control	0.03	−3.90
Moryella	S vs Control	0.04	−4.25
Peptostreptococcus	L1 vs Control	0.04	1.90
Rothia	S vs Control	0.002	−1.96
	L1 vs Control	0.005	−2.98
Selenomonas	L0 vs Control	0.001	3.07
Staphylococcus	L0 vs Control	0.01	3.11
Stenotrophomonas	L0 vs Control	0.03	4.26
Veillonella	S vs Control	0.01	−1.21
	L1 vs L0	0.003	−1.67
	L1 vs Control	0.005	−2.22

S: OSCC; L0: OSCC without active lesion; L1: OSCC with active lesion; healthy (C): individuals without disease.

cancer samples (Fig. 2C). Taking into account the relative distribution of these 15 differentially abundant taxonomical groups, it was possible to depict microorganisms that were under-represented in L1 (C or L0 > L1) and over-represented in L1 (L1 > C and/or L0) (Fig. 2D).

Compared to control individuals, the abundances of *Abiotrophia*, *Acinetobacter*, *Alloscordovia*, *Dialister*, *Gemella*, *Granulicatella*, *Peptoestreptococcus*, *Selenomonas*, *Staphylococcus,* and *Stenotrophomonas* were considerably higher in L0 and L1 patients (Table 1, Student's *t*-test, p < 0.05). On the other hand, the abundances of *Veillonella*, *Rothia*, *Moryella*, *Kingella,* and *Centipeda* were lower in L0 and L1 patients compared to control individuals (Table 1, Student's *t*-test, p < 0.05). In addition, L1 patients exhibited greater levels of *Alloscordovia* and reduced abundances of *Veillonella* when compared to L0 patients (Table 1, Student's *t*-test, p < 0.05).

We next inferred the bacterial sub-pathways using the Burrito interactive tool to examine differential patterns of microbial functions in control individuals and L0 and L1 patients (Supplementary Table 16, Supplementary Fig. 1). Interestingly, we observed 11 significant bacterial sub-pathways distributed among the sample groups (control vs L0, control vs L1, and L0 vs L1; Student's t-test, P-value < 0.05). Compared to control individuals, L1 patients displayed an increase in prokaryotic defense system maintenance and thiamine metabolism, while L0 patients displayed an increase in flagellar assembly and butanoate metabolism. Lysine biosynthesis, pantothenate, and CoA biosynthesis as well as signal transduction mechanisms are under-represented in L1 patients. However, focusing on the differences between OSCC patients with and without active lesions (L0 vs L1), lipid biosynthesis proteins and thiamine metabolism are shown to be under-represented in L1, while nicotinate and nicotinamide metabolism, pentose phosphate pathways, and signal transduction mechanisms are over-represented. Taken together, the results suggest that besides the involvement of different taxonomy compositions in L0 and L1 patients compared to control individuals, the bacterial genetic repertoire and thus, the microbiome structure and function as well can be changed in OSCC patients in both stages (before and after surgical treatment) (Fig. 2E).

3.3. Changes in oral microbiota are associated with prognostic factors and poor overall patient survival

To evaluate whether the relative abundance of a given microorganism in saliva could reflect important clinical characteristics, we first associated taxa genus abundance with the clinical data. We observed that seven different genera (*Gemella*, *Granulicatella*, *Veillonella*, *Acitenobacter*, *Centipedia*, *Selenomonas,* and *Staphylococcus*) correlated significantly (P-value < 0.05) with clinical data related to OSCC prognosis (e.g., N/Nlevel, lesion, treatment, clinical stage, etc.). The relative abundance of *Veillonella* and *Centipeda* had a stronger correlation (Table 2, linear regression, P-value < 0.05, $R < -0.6$, or $R > 0.6$) with clinical tumor size (cT), lesion, and clinical stage (Fig. 3A). On the other hand, only three genera (*Dialister*, *Rothia,* and *Peptoestreptococcus*) correlated with the cluster formed in hierarchical clustering (Table 2, linear regression, P-value < 0.05, $R < -0.5$ or $R > 0.5$). However, cluster formation did not significantly correlate with any clinical feature under Pearson Chi-Square correlation. In addition, the relative abundances of six genera (*Stenotrophomonas*, *Staphylococcus*, *Selenomonas*, *Centipeda*, *Alloscardovia*, and *Acinetobacter*) are associated with overall patient survival, as suggested by an analysis of Kaplan-Meier survival curves (Fig. 3B and Supplementary Table 14). In all cases, the higher expression of the genus is associated with a decrease in overall survival in accordance with its classification as over-represented in L1 and the microorganisms that are potentially harmful for OSCC (Fig. 2D). Furthermore, ROC curves were used to evaluate the power of the 15 different genera to distinguish control, L0, and L1 patients based on their distribution profiles (Supplementary Fig. 5 and Supplementary Table 15). ROC analysis has shown that the abundances of different genera obtained by 16S rDNA sequencing allow us to distinguish C from L1, L0 from L1, and C from L0 patients with a significant P-value (<0.05) and an AUC above 0.75 (Supplementary Table 15). Among the microorganisms whose

Table 2
Linear regression with Pearson Correlation coefficient determination between i) Genus taxonomy relative expression and clinicopathological data and ii) Genus taxonomy relative expression and cluster formation.

Microorganism (genus taxonomy)	Clinicopathological feature/Cluster (N)	Linear regression P-value*	Correlation coefficient (*R*)**
Centipeda	Lesion (10)	0.0202	**0.7148**
Veillonella	Lesion (15)	0.0039	**−0.6965**
	cT (15)	0.0141	−0.6178
	pTumor size (15)	0.0141	−0.6178
	Clinical stage (15)	0.0076	−0.6584
	Site (15)	0.0012	**−0.7537**
Acinetobacter	Treatment (8)	0.0296	**−0.7573**
Dialister	Cluster (15)	0.021	0.5884
	Cluster (21)	0.0015	0.6243
Granulicatella	cN_grouped (15)	0.0498	−0.5144
	Lesion (15)	0.0067	−0.5488
Gemella	Clinical stage (15)	0.0462	0.5214
Selenomonas	Treatment (15)	0.0207	−0.5895
Staphylococcus	Treatment (14)	0.0202	−0.6112
Peptostreptococcus	Cluster (13)	0.0189	0.5326
Rothia	Cluster (15)	0.0088	−0.5334

The significant correlations are listed in the table and the stronger associations ($-0.7 > R > 0.7$) are highlighted in bold. Clinical features: Age – patients divided by the average age of the cohort (57 years); Lesion - Control, L0 and L1; Site - Tongue, mouth floor, soft palate or other sites; cT - 1, 2, 3 or 4; cT_grouped - 1-2, 3-4; pTumor size - 1, 2, 3 or 4; pT_grouped - 1-2, 3-4; cN - 1, 2 or 3; cN_grouped - 1-2, 3; pN - 1, 2 or 3; pN_grouped - 1-2, 3; Gender – Masculine or Feminine; Clinical stage-1- T1 N0 M0, 2- T2 N0 M0, 3- T1-2 N1 M0 or T3 N0-1 M0, 4- T1-3 N2 M0 or T4a N0-2 M0, 5- T N3 M0 or T4b any N M0; Treatment-1-no treatment or refused treatment, 2-surgery, 3-combined treatment. Cluster with 3 groups was formed using Euclidean Ward D Method and considering all 15 differential taxonomies at the genus level.

* Significance threshold P-value ≤ 0.05.

** Correlation coefficient $-0.5 > R > 0.5$ and Multiple $R^2 > 0.3$.

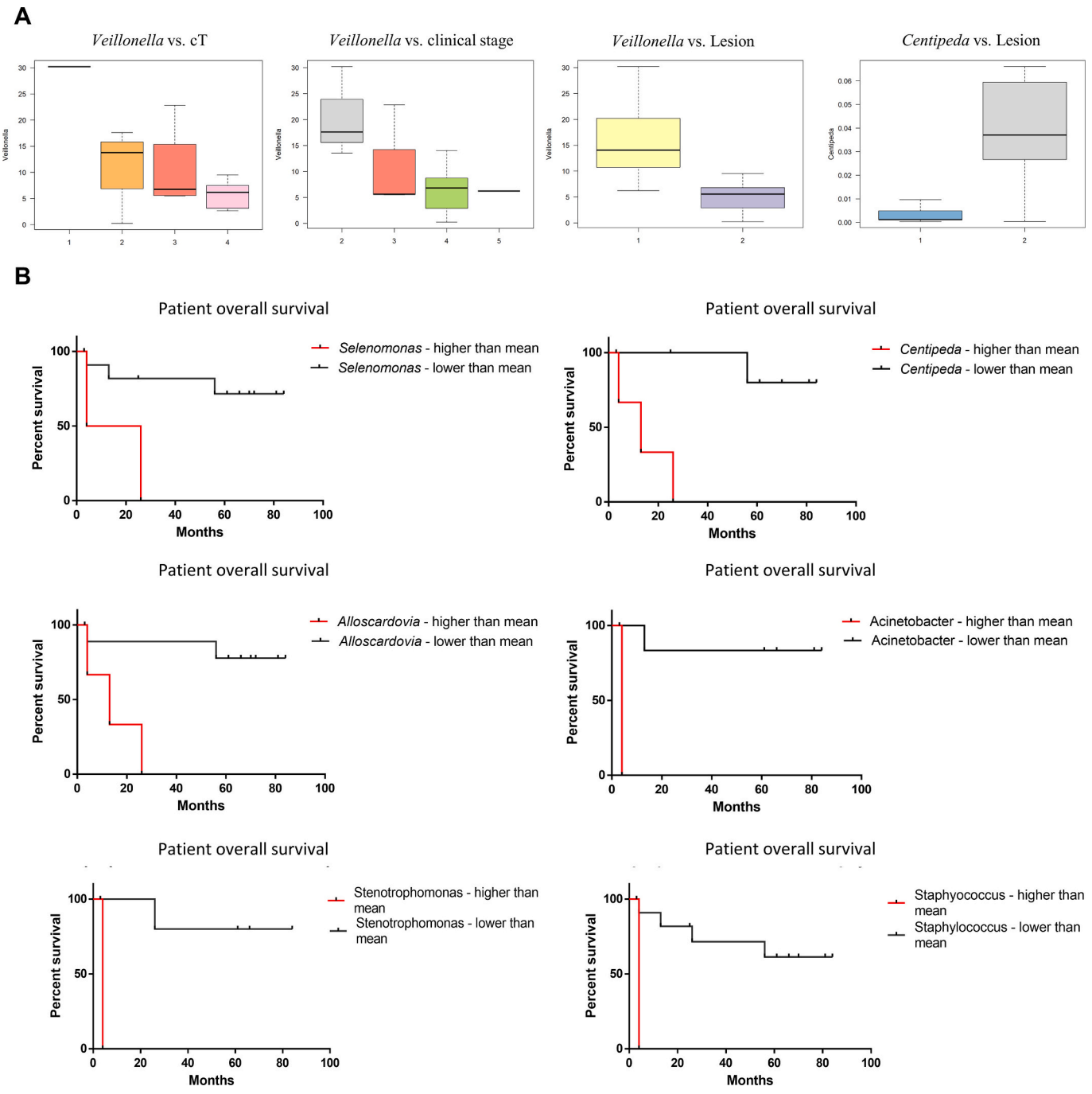
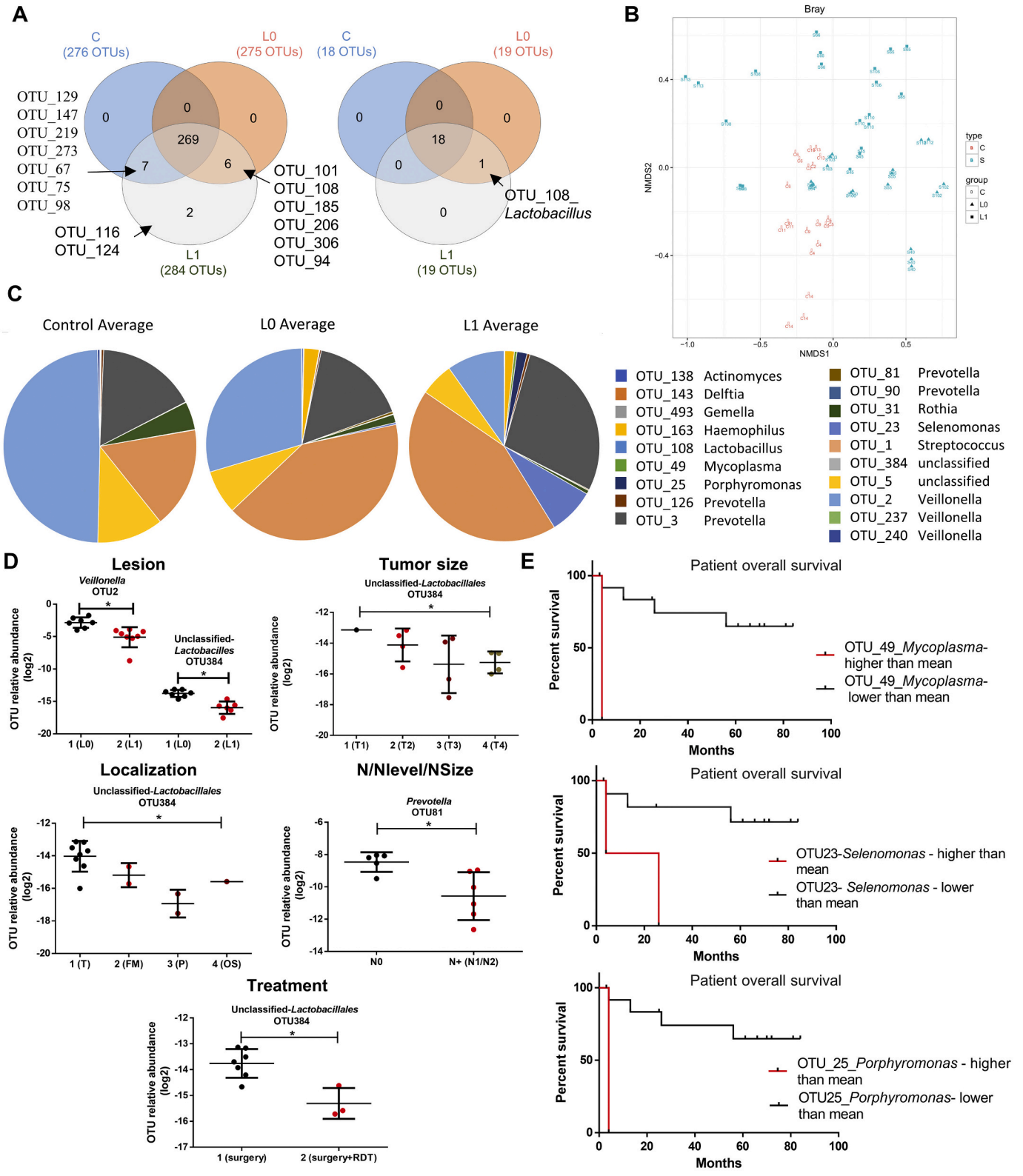


Fig. 3. Correlation and Kaplan – Meier survival analysis of genera relative abundance with clinical factors and clinical outcomes considering Overall survival (OS). (A) Among the differential genera, the taxonomy distribution of *Veillonella* and *Centipeda* correlates with the presence of active lesion. Also, considering the clinicopathological characteristics associated to OSCC progression, *Veillonella* relative abundance inversely correlates with clinical tumor size (cT; 1: T1, 2:T2, 3:T3, 4:T4), lesion (1, no active lesion 2, with active lesion) and clinical stage (2, T1 N0 M0, 3, T2 N0 M0, 4: T1–-2 N1 M0 and T3 N0-1 M0, 5: T1-3 N2 M0 and T4a N0-2 M0), suggesting that a higher abundance of *Veillonella* in saliva can implicate in a better prognosis for OSCC patients. (B) Higher *Centipeda*, *Selenomonas*, *Alloscardovia*, *Acinetobacter*, *Stenotrophomonas* and *Staphylococcus* abundance in the saliva had patient worse survival outcome (P-value < 0.05, log-rank test).

identification could be used for OSCC diagnostic classification, the results highlighted *Veillonella*, *Rothia*, and *Granulicatella* (Supplementary Fig. 5 and Supplementary Table 15). These results suggest that the relative abundance of these microorganisms can not only be associated with clinicopathological data and prognosis value (tumor size aggressiveness and clinical stage) but can also be used for sample group classification.

3.4. Targeted strategy to analyze the genera selected by feature analysis

Since our main goal was to identify the clinical relevance of changes in the microbiome and select the best genera for a proteomic database search, we performed a feature selection analysis to reduce the OTUs scope while maximizing group separation. This analysis resulted in the selection of 19 different OTUs (Fig. 4A) that were still able to discretize the samples (Fig. 4B, Supplementary Fig. 2 and Supplementary Table 5–6).



(caption on next page)

Fig. 4. Microbiome saliva profiling by feature-selected OTUs analysis can better separate the samples in three groups and is associated with presence of active lesion. (A) Distribution of OTUs identified by 16S analysis (left side) and the reduced feature-selected OTUs (right side). The OTUs described as exclusive in certain sample groups correspond to the following genus levels: Unclassified (*Clostridiales_Incertae Sedis XIII)*_OTU129, *Catonella*_OTU147, Unclassified (*Neisseriaceae)*_OTU219, Unclassified (*Bacteroidetes*)_OTU273, Unclassified (Bacteria)_OTU67, *Prevotella*_OTU75, *Prevotella*_OTU98 (common among Control and L1 patients); *Prevotella*_OTU116, *Gardnerella*_OTU124 (Exclusive in L1 patients); Unclassified (*Veillonellaceae)*_OTU101, *Lactobacillus*_OTU108, *Bradyrhizobium*_OTU185, Unclassified (*Flavobacteriaceae*)_OTU206, *Pelomonas*_OTU306, *Rubrobacter*_OTU_94 (common among L0 and L1 patients). (B) *Nonmetric multidimensional scaling* analysis *(NMDS)* using Bray Curtis distance considering the reduced feature-selected OTUs data. (C) Microbiome distribution was determined considering the average value of each OTU in saliva from control individuals and OSCC patients (*with* and *without* active lesions, L0 and L1, respectively). It is shown that the most frequent genera in control individuals is *Veillonella*, while in L0 patients it is *Veillonella* and *Streptococcus* and in L1 patients it is *Streptococcus* only. The OTUs described in the Venn diagram correspond to the following genus levels: *Prevotella*_OTU90; *Prevotella*_OTU126; *Prevotella*_OTU3; *Prevotella*_OTU81; *Porphyromonas*_OTU25; *Rothia*_OTU31; *Actinomyces*_OTU138; Unclassified_OTU5; *Haemophilus*_OTU163; *Delftia*_OTU143; *Veillonella*_OTU2; *Veillonella*_OTU240; *Veillonella*_OTU237; *Selenomonas*_OTU23; *Lactobacillus*_OTU108; *Gemella*_OTU493; Unclassified_OTU384; *Streptococcus*_OTU1; *Mycoplasma*_OTU49. (D) Correlation of OTU abundance with OSCC clinical factors. The abundance of 3 different OTUs in OSCC saliva were correlated with clinical features such as presence of active lesion (1:L0; 2:L1), tumor size (1,T1, 2:T2, 3:T3, 4:T4) and site (1, T-tongue; 2:FM-floor of the mouth; 3: P-palate; 4:OS-other sites), lymph node metastasis (1, N0; 2: N + -N1/N2) and treatment (1-surgery, 2-surgery+ RT, radiotherapy). Significance (*) defined by Mann-Whitney or Kruskal Wallis *P*-value ≤0.05; minimum correlation coefficient + 0.7/−0.7; multiple $R^2 >$ 0.5. (E) Higher *Selenomonas*, *Porphyromonas*, and *Mycoplasma* abundance in the saliva led to patient higher risk of worse survival (P-value < 0.05, log-rank test).

To compare the composition of the microbial community within the oral cavity, the presence of bacterial genera as well as their relative abundance within the sample groups was taken into account. Using the relative abundance of the OTUs, it is possible to show a difference in their distribution among the sample groups (Figs. 4C). In order to evaluate whether the relative abundance of a microorganism in saliva could reveal important clinical characteristics, we first associated OTU profile abundance with 27 different clinical data—including tumor-node-metastasis (TNM) classification [57], which includes active lesion, tumor site, and size (Fig. 4D)—and we observed that the relative abundance of *Veillonella* (OTU2) and unclassified *Lactobacillales* (OTU384) correlates with lesion state (L0 or L1), while unclassified *Lactobacillales* (OTU384) correlates with tumor size, localization and treatment, and *Prevotella*'s (OTU2) relative abundance correlates with local metastasis presence (N), size, and level (N1/N2 vs N3/N4) (Pearson correlation >0.7, Multiple $R^2 > 0.5$, and linear regression followed by Mann-Whitney test; P-value < 0.05, Table 3). Moreover, the relative abundances of three OTUs (*Selenomonas*, *Porphyromonas*, and *Mycoplasma*) were associated with overall patient survival, as suggested by the analysis of Kaplan-Meier survival curves (Fig. 4E and Supplementary Table 14).

3.5. Saliva metaproteomics shows that host and microbiome protein expression is not recovered after surgery treatment

Aiming to gain further insights into functional changes in the oral microbiota associated with OSCC and to identify the microbial protein composition of OSCC-derived saliva proteins taken from saliva cells and supernatant, we determined the metaproteomic profiles of the 24 samples (C, L0, and L1, Supplementary Table 1) using LC-MS/MS. For this, the 12 different genera from the 19 feature-selected OTUs (*Streptococcus*, *Lactobacillus*, *Prevotella*, *Actinomyces*, *Delftia*, *Haemophilus*, *Veillonella*, *Selenomonas*, *Porphyromonas*, *Rothia*, *Mycoplasma*, *Gemella*) were studied in this analysis (Fig. 4A-B; Supplementary Fig. 2; and Supplementary Table 5–6).

As known from previous reports, a microbial peptide often matches to multiple proteins with high sequence similarity, resulting in difficulties in identifying the microbial origin of the individual peptides. To alleviate ambiguities, we applied two different strategies: (i) we performed a database search enabling a division by taxonomy with MaxQuant (named Database 1), and (ii) we performed a database search with a non-redundant meta-protein database constructed to contain only unique microbial peptides at the genus level (named Database 2). Both meta-databases were merged with the human proteome. As a result, we identified a total of 187 (25%) microbial proteins and 569 (75%) human proteins (Supplementary Tables 9–12) in the saliva supernatant and cells using both databases (redundant and non-redundant). The intensities of these unique and razor peptides were further used for meta-protein quantification (Supplementary Table 9–12).

Table 3
Linear regression with Pearson Correlation coefficient determination between OTUs relative abundance and clinicopathological data.

Microorganism (genus taxonomy)	Clinicopathological status/Cluster (N)	Linear regression P-value*	Correlation coefficient (*R*)**
*Actinomyces*_OTU138	cN_grouped	0.0064	**−0.6688**
*Veillonella*_OTU2	Lesion	0.0017	**−0.7371**
	cT	0.0092	−0.6463
	pTumor size	0.0092	−0.6463
	Clinical stage	0.0084	−0.6524
	Site	0.0032	**−0.7071**
*Veillonella*_OTU240	Lesion	0.0034	**−0.704**
	cT_grouped	0.0077	−0.6577
	pTumorSize_grouped	0.0077	−0.6577
	pTumor size	0.0034	**−0.704**
	Site	0.0122	−0.6282
Unclassified_OTU384 (*Lactobacillales*)	Lesion	1.00E-04	**−0.8493**
	cT	0.0032	**−0.7078**
	cT_grouped	0.0126	−0.6254
	pTumorSize_grouped	0.0126	−0.6254
	pTumor size	0.0032	**−0.7078**
	Clinical stage	0.0114	−0.6324
	Site	0.0021	**−0.7276**
*Prevotella*_OTU81	cN	0.0033	**−0.7049**
	cN_grouped	2.00E-04	**−0.8205**
	pN Size	0.0103	−0.6394
	pN	3.00E-04	**−0.8096**
	pNlevelt	3.00E-04	**−0.8096**

The significant correlations are listed in the table and the stronger associations (−0.7 > R > 0.7) are highlighted in bold. Clinical features: Age – patients divided by the average age of the cohort (57 years); Lesion - Control, L0 and L1; Site - Tongue, mouth floor, soft palate or other sites; cT - 1, 2, 3 or 4; cT_grouped - 1-2, 3-4; pTumor size - 1, 2, 3 or 4; pT_grouped - 1-2, 3-4; cN - 1, 2 or 3; cN_grouped - 1-2, 3; pN - 1, 2 or 3; pN_grouped - 1-2, 3; Gender – Masculine or Feminine; Clinical stage-1- T1 N0 M0, 2- T2 N0 M0, 3- T1-2 N1 M0 or T3 N0-1 M0, 4- T1-3 N2 M0 or T4a N0-2 M0, 5- T N3 M0 or T4b any N M0; Treatment-1-no treatment or refused treatment, 2-surgery, 3-combined treatment. Cluster with 3 groups was formed using Euclidean Ward D Method and considering all 15 differential taxonomies at the genus level.

* Significance threshold P-value ≤ 0.05.

** Correlation coefficient − 0.5 > R > 0.5 and Multiple $R^2 > 0.39$.

The filtered dataset from the supernatant samples resulted in 205, 185, and 176 proteins identified for control, L0, and L1, respectively, in Database 1 (Human_HOMD_16S_complete), and 166, 135, and 128 proteins for control, L0, and L1, respectively (Fig. 5A), in Database 2 (Human_HOMD_16S_unique). To evaluate the contribution of bacterial protein identification to the supernatant saliva proteome, the abundance and frequency of the human and bacterial proteins were plotted together, integrating the results from both databases. The protein distribution and the dynamic range indicated that the proteins that belonged to the high abundance range were mostly human (Fig. 5B).

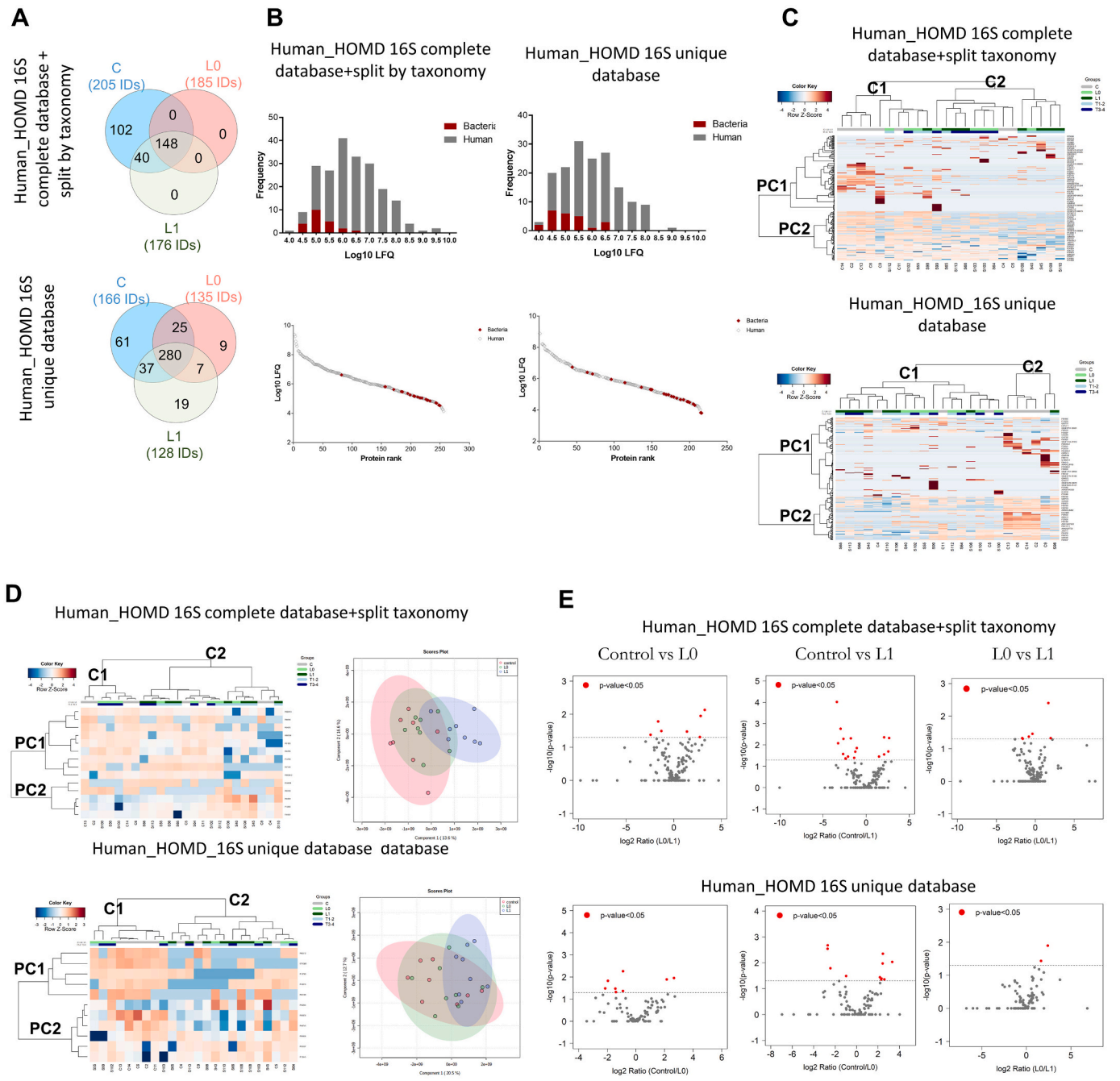


Fig. 5. Metaproteomics of saliva supernatant shows the contribution of the microbiome proteins is lower than human proteins. (A) Venn diagram of saliva supernatant-derived proteins exclusive or commonly identified among the sample groups using two distinct databases. (B) The contribution of 270 human and 50 bacterial proteins identified in saliva samples using database 1 (HOMD 16S_complete) or database 2 (HOMD 16S_unique) merged with UniProt human database is displayed as histogram of frequency intensity (log10LFQ) distribution in the upper panel, and the dynamic range of human and bacterial proteins ranked by their intensities (log10LFQ) in the lower panel. (C) Heat map of the total cellular saliva proteome of both databases and (D) differential proteins (ANOVA test, *P*-value<0.05) shows no clear grouping of samples based on a single classifier feature (presence or absence of oral cancer). Hierarchical clustering was performed in R environment and Caberra Ward D method for database 1 (Human merged with HOMD_16S complete database) search (shown above) and database 2 (Human merged with HOMD_16S unique database) search (shown below) was applied. Partial least squares discriminant analysis (PLS-DA) demonstrates that cellular protein composition in saliva of Control individuals and OSCC patients (L0 + L1) cannot be distinguished when using the non-redundant database (upper) or redundant database (lower). (E) Differential proteins quantified in Control and L0, Control and L1, and L0 and L1 samples are represented on a volcano plot. Proteins above the significance threshold (Student's *t*-test P-value ≤0.05) are highlighted as red dots (up or downregulated).

Also, the proportional abundance of human proteins in relation to bacteria proteins was not different among the sample groups (Supplementary Fig. 4), although it is possible to notice a small increase in bacterial abundance in the L0 samples. Furthermore, hierarchical clustering did not separate samples into two or three distinct groups (Fig. 5C–D). Heat map clustering analyses indicated that the proteomic data from supernatant saliva samples were less effective in their ability to classify samples compared to 16S rDNA. The volcano plot suggests significant proteins between control and L0 and control and L1, although no significant proteins came from bacterial taxonomy (Fig. 5E). Moreover, the protein distribution profile between L0 and L1 did not differ, suggesting that there were no changes in protein

abundance between these sample groups.

The filtered dataset from the saliva cells yielded 290, 148, and 188 proteins from control, L0, and L1, respectively, in Database 1 (Human_HOMD_16S_complete) and 403, 331, and 353 proteins in control, L0, and L1, respectively (Fig. 6A), in Database 2 (Human_HOMD_16S_unique). To evaluate the contribution of bacterial protein identification to the cellular saliva proteome, the abundance and frequency of the human and bacterial proteins were compared, revealing that the proteins with higher abundance were mostly of human origin (Fig. 6B). Although the contribution of bacterial proteome in saliva cells is more frequent when compared to the supernatant dataset (Fig. 5B), the proportional abundance of human proteins in relation to bacterial proteins was not different among the sample groups (Supplementary Fig. 4). Furthermore, the identified proteins were not able to separate the samples into three distinct groups by hierarchical clustering (Fig. 6C–D), regardless of database and input dataset. Heat map clusters

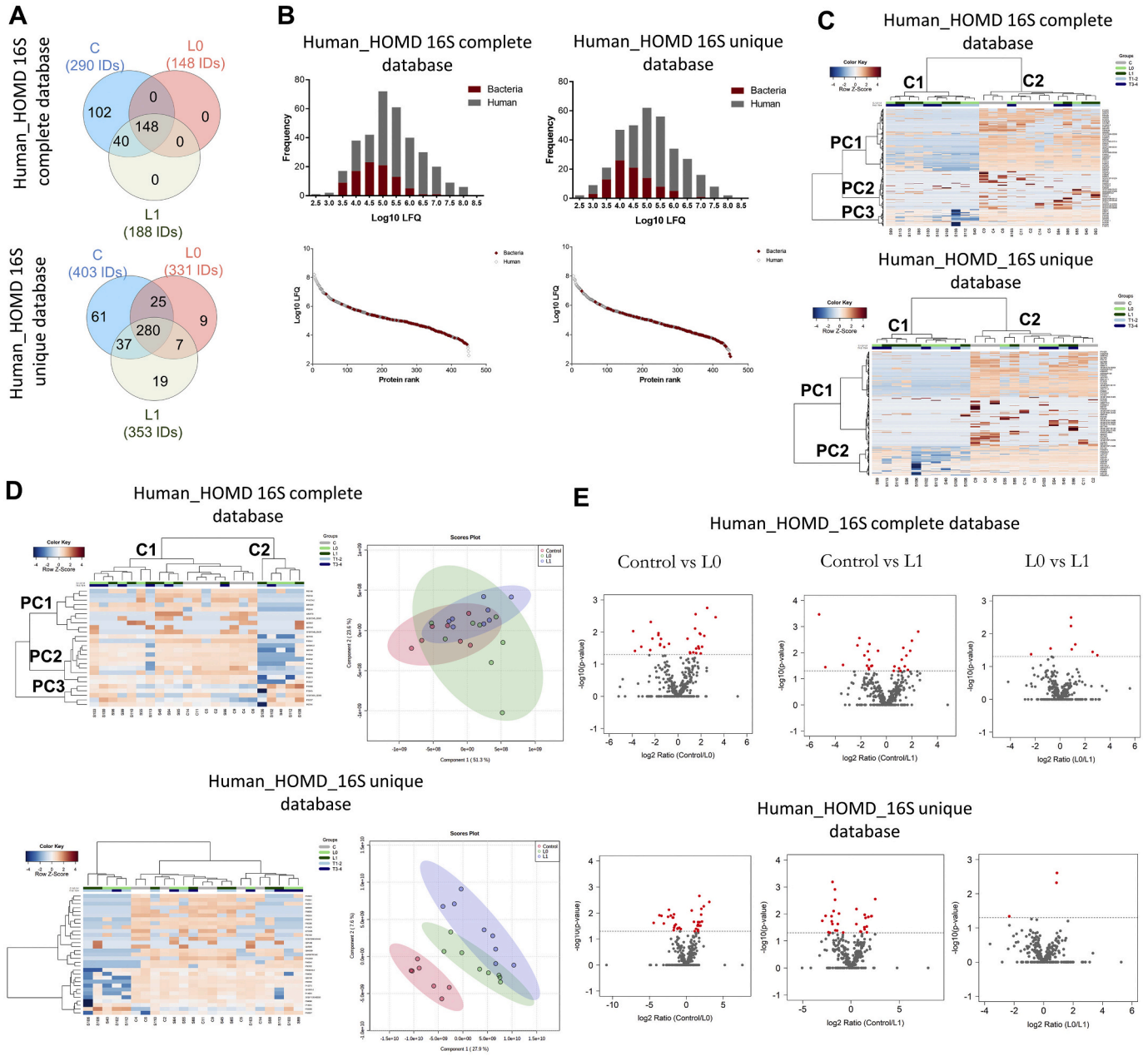


Fig. 6. Metaprotemics in saliva cells shows the contribution of the microbiome proteins is lower than human proteins. (A) Venn diagram of saliva cell-derived proteins exclusive or commonly identified among sample groups using two distinct databases. (B) The contribution of 350 human and 100 bacterial proteins identified in saliva samples using database 1 (HOMD 16S_complete) or database 2 (HOMD 16S_unique) merged with UniProt human database is displayed as a histogram of frequency intensity (log10LFQ) distribution in the upper panel, and the dynamic range of human and bacterial proteins ranked by their intensities (log10LFQ) in the lower panel. (C) Heat map of the total cellular saliva proteome of both databases and (D) differential proteins (ANOVA test, P-value<0.05) shows no clear grouping of samples based on a single classifier feature (presence or absence of oral cancer). Hierarchical clustering was performed in R environment and Caberra Ward D method for database 1 (Human merged with HOMD_16S complete database) search (shown above) and database 2 (Human merged with HOMD_16S unique database) search (shown below) was applied. Partial least squares discriminant analysis (PLS-DA) demonstrates that cellular protein composition in saliva of Control individuals and OSCC patients (L0 + L1) cannot be distinguished when using the non-redundant database (upper) or redundant database (lower). (E) Differential proteins quantified in Control and L0, Control and L1, and L0 and L1 samples are represented on a volcano plot. Proteins above the significance threshold (Student's *t*-test, P-value $\leq$0.05) are highlighted as red dots (up or downregulated).

indicated that the proteomic data from saliva cell samples also produced a less accurate classification of the sample groups compared to the 16S rDNA data, suggesting that this proteomic data is more heterogenous. On the other hand, a PLS-DA plot with both databases (Fig. 6E) was able to partially classify the saliva into three distinct groups only with the non-redundant database, and again, L0 exhibited a more similar profile to L1 than to the control samples. Moreover, the volcano plot showed significant human and bacterial proteins between control and L0 and control and L1, while the protein profile distribution did not differ between L0 and L1 with changes in protein abundance and P-value. This indicates that although the active lesion was removed, the protein composition in the cellular saliva reflected and maintained the microenvironment change promoted by the tumor development (Fig. 6F).

In addition, among the 350 detected human proteins in cellular and supernatant saliva, we identified 44 human proteins that were shared among the sample groups, and are statistically significant (Student's *t*-test, P-value<0.05) (Supplementary Fig. 5A). Eleven of the differential human proteins (Cathepsin G (CATG), Histidine-rich glycoprotein (HRGP), BPI fold-containing family B member 1 and 2 (BPIFA1 and 2), Protein S100-A9 and A8 (S100A9 and S100A8), Glyceraldehyde-3 phosphate dehydrogenase (GAPDH or G3P), Lactoferrin (LTF), Immunoglobulin heavy constant alpha 1 (IGHA1), Neutrophil gelatinase-associated lipocalin (NGAL), Neutrophil elastase (ELNE)) participate in the protein-protein interaction network constructed using STRING, that also showed differences in the interacting network when comparing the differential proteins given by distinct sample groups (Control, L0 and L1) (Supplementary Fig. 5B). Only LTF and NGAL abundance correlated with clinicopathological data ($R > 0.5$, Linear regression, $p < 0.05$ and $R < -0.5$ or $R > 0.5$, Supplementary Table 13). We show that the abundance of these proteins is increased in OSCC patients compared to Control and four of the differential proteins (S100-A9, ELNE, CTSG, BPIB1) were also able to distinguish the sample groups (Supplementary Fig. 6A-B). Finally, we performed a combination of MG data (relative quantification of *Granulicatella*, *Selenomonas*, *Veillonella*, and *Rothia*) with proteomic data (S100A-9, ELNE, CTSG, BPIB1) that showed best AUC ROC values (Supplementary Fig. 6C). Although for Control vs L1 the combination of the data was able to distinguish the groups with an AUC value close to 0.8, indicating a promising microbiome-host signature for differentiating OSCC from Control individuals, the other comparisons (Control vs L0 and L0 vs L1) did not present the same positive scenario.

3.6. Metagenomes and metaproteomes in saliva reveal complementary features regarding of microbial composition and its functional characteristics

Comparing the microbial composition of DNA (MG) and proteins (MP) from the oral cavities of different patients, we took into account the annotated microbial compositions and functional features. Focusing on the main assignments obtained by the metagenomic data, a prevalence of *Firmicutes* (51%) was revealed, followed by *Bacteroidetes* (22%) and *Proteobacteria* (10%). Considering all the bacterial proteins identified in the cellular component of saliva with Database 2 (Human_HOMD_16S unique database), taxonomic annotation at the phylum level revealed a prevalence of *Firmicutes* (71%), followed by *Actinobacteria* (13%) and *Bacteroidetes* (10%) (Fig. 7A). Although there is an overall consistency, a significantly higher percentage of annotated proteins being assigned to *Firmicutes* and *Actinobacteria* suggests that they might display an overall higher protein production than the other phyla across the 24 samples.

In addition, we determined the relationship between the number of meta-proteins and the abundances at the genus level based on metagenomics. Considering that the more abundant a given genus was (based on metagenomics analysis), the more of the identified metaproteins should be assigned to this genus, we showed that *Streptococcus*, *Prevotella*, and *Veillonella* harbored the most assigned meta-proteins and were among the most representative genera in metagenomics as well (Fig. 7B). Comparing KEGG functional categories based on metagenomics and metaproteomics data, we also observed concordance in the relative contribution of individual categories between the two datasets. For instance, as determined by metaproteomics, 40% and 35% of the proteins were assigned to carbohydrate metabolism and genetic information processing, whereas the corresponding metagenomic percentages of both categories were 50% and 40%, respectively (Fig. 7C). We found that 82 meta-proteins, accounting for 80% of all identified meta-proteins, could be assigned to 20 KEGG orthologues (KO) (Fig. 7D). The top KOs harbored 16 proteins, annotated as genetic information processing-plant pathogen interaction pathways (K02358), carbohydrate metabolism-glycolysis pathways (K01689), and genetic information processing-ribosome pathways (K02935) (Fig. 7E). Taking into account the percentage of bacterial proteins annotated in each process, genetic information processing (25%) produces a higher percentage of annotated meta-proteins. Furthermore, hierarchical clustering of the bacterial abundances (LFQ expression) associated with different KEGG processes indicated the formation of three distinct clusters (Fig. 7F). Aiming to link the microbial protein patterns to metagenomic microbial abundances, we next conducted a distribution analysis of meta-proteins. Showing disagreement with our metagenomic findings (Fig. 4C), the levels of *Prevotella* meta-proteins tended to be lower in L0 and L1 patients (Fig. 7F), although *Prevotella*, as annotated in the metagenome data, exhibited significantly higher abundance in these patients.

4. Discussion

In a healthy state, the microbiome interacts synergistically with the host and contributes to normal tissue functions while protecting the host from pathogenic colonization. Alterations in the composition of the microbiome, also known as dysbiosis, are commonly associated with dysregulated immune responses and have been shown to be involved in various disease states, including inflammatory diseases of the digestive tract [58] and diseases of the oral cavity, such as periodontitis and oral cancer [59,60]. Although, the importance of the microbiome in health and its dysbiosis has been elucidated in the past years, the multitude of host and environmental factors that influence the microbiome are still largely unknown. To identify microbiome changes and to correlate them with clinical data relevant for diagnosis and/or prognosis of OSCC patients, we first characterized the metagenomic and metaproteomic profiling of saliva from 24 patients (8C, 8 L0, and 8 L1) using 16S rDNA sequencing and mass spectrometry (Fig. 1).

First, we mapped the metagenomic profile of these three groups, which contributes to a better understanding of the microbiome composition in control individuals and OSCC patients. It is important to reinforce that we have characterized only bacteria and that other components of the oral microbiome that can play a role in OSCC such as viruses, fungi, and others, have not been analyzed in this study. We identified 68 taxonomies common to all groups (Fig. 2A), and the most frequent identifications at the genus level were *Streptococcus*, *Veillonella* and *Prevotella* (Fig. 2B). We found dramatic changes in the bacterial taxa's composition and function in all different conditions (C, L0, and L1) (Fig. 2C-D). Significantly different abundances of 15 distinct taxonomies were shown between C, L0, and L1 (Fig. 2D). In addition, we also found changes in the oral microbiome after surgical removal in L0 patients when compared to those with an active lesion (L1) (Fig. 2). With this approach, we found that distinct annotated bacterial biological processes were overrepresented in L1 compared to C or L0, such as the prokaryote defense system and nicotinate and nicotinamide metabolism (Fig. 2E). Interestingly, altered metabolic needs, including an accelerated rate of nicotinamide adenine dinucleotide (NAD^+) cycling, is one of the alterations characterized in cancer cells compared to normal cells [61]. Additionally, the increase in the prokaryote response system may suggest that L1 patients have triggered reactions to activate their

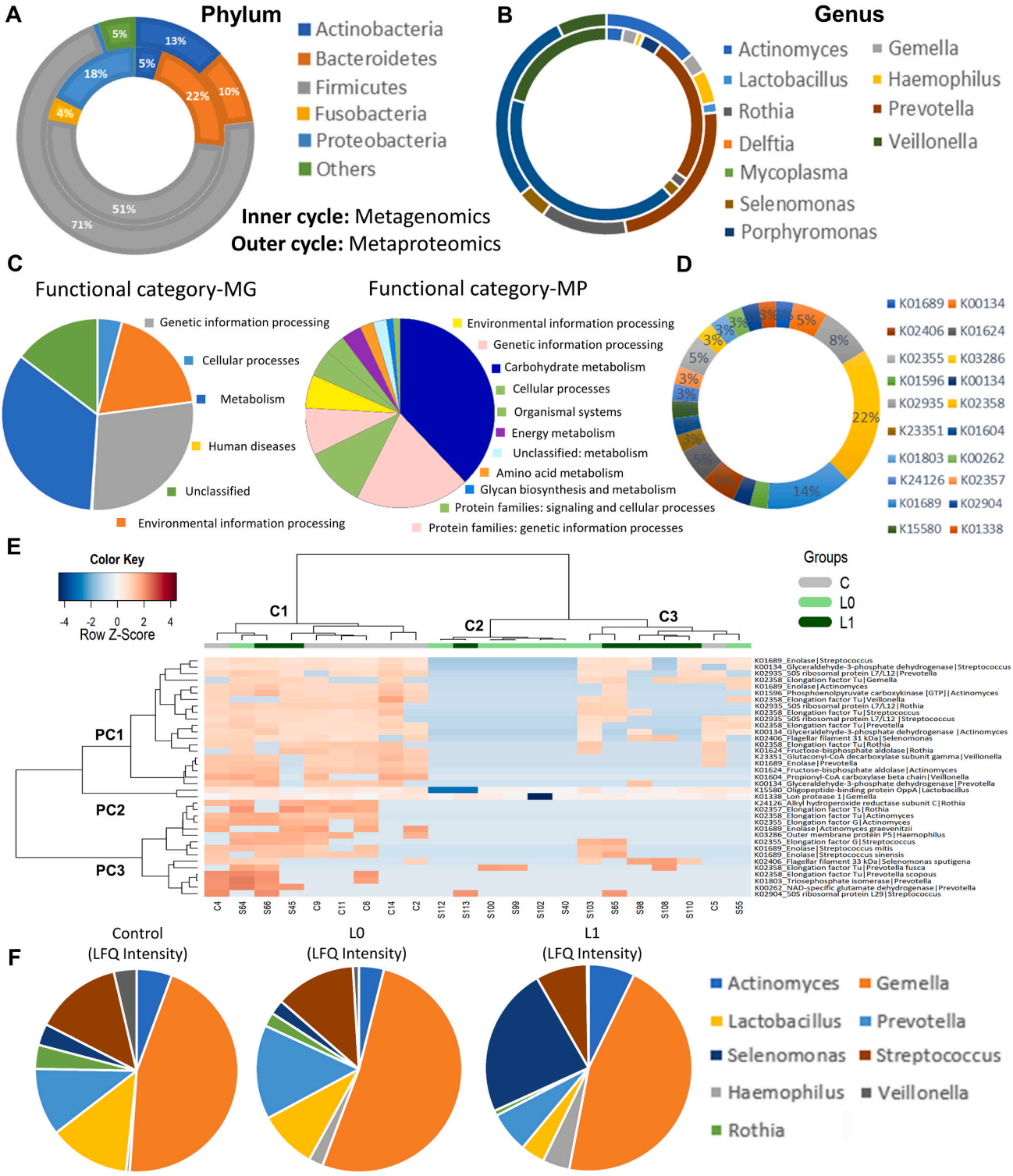


Fig. 7. Metaproteomics data from saliva cell samples is able to distinguish control individuals from OSCC patients. A) Taxonomical Phylum characterization of all bacterial proteins identified in saliva cells from proteomics analysis using database 2 (Human merged HOMD_16S_unique_database). B) The microbiome protein distribution in saliva from control individuals and OSCC patients (without and with active lesions, L0 and L1, respectively) can be seen visualized in the pie chart and when compared to the contribution of each genus identified by the feature-selected 19 OTUs, the distribution is not alike. The relative abundance of genus related proteins was calculated by summing the number of identified proteins related to the same taxonomy at the genus level. C) Functional characterization of all bacterial proteins identified in saliva cells with database 2 (Human merged HOMD_16S_unique_database) and 16S rDNA relative abundance using KEGG GhostKOALA and Burrito, respectively. D) The number of bacterial proteins assigned to each functional process is shown in percentage values. E) Heatmap shows the LFQ intensity for all samples in each subgroup. Each row represents one protein group, and the corresponding KO id, KO name and taxonomic assignment were indicated in the right panel. KO name and taxonomic assignment were obtained using KEGG GhostKOALA database. The hierarchical clustering was generated in R environment using the Ward D method. F) Considering the genus level of all bacterial proteins identified by MP, the distribution is different among the sample groups but does not match the MG data.

defense system against the enhanced relative abundances of Proteobacteria and Firmicutes.

Corroborating previous results in the literature, our results also show that *Veillonella* and *Granulicatella* are abundant oral microorganisms in healthy individuals and may have beneficial effects on the microenvironment. Similar to what has been shown in other studies, *Centipeda* was found to be more abundant in OSCC patients with tumor lesions (L1) in our data [59,62] (Fig. 2D). Still differences in microbiome studies have been seen and possible causes include the differences in sample collection, stratification of samples, sample size, and miSeq data collection and analysis [59,62]. Our results also show that the abundance of *Centipeda* and *Veillonella* was correlated with the presence of active lesions, and *Veillonella* also correlated with other clinical factors associated with OSCC prognosis, such as clinical (cT) and pathological tumor size (pT) and clinical stage. *Granulicatella* was the only differential microorganism correlated with lymph node metastasis (cN grouped), one of the main prognostic factors for OSCC (Fig. 3A). In addition, high *Centipeda* levels were correlated with a shorter overall survival rate in our data together with other taxonomies that may contribute to harmful effects on the microenvironment (Fig. 3B).

After identifying the significant taxonomies from different sample groups, we subsequently verified the ability of the 16S rDNA abundance to distinguish patients based on the presence (L1) or absence of an active lesion (C or L0) (Fig. 4). *Rothia*, *Veillonella*, *Staphylococcus*, *Centipeda*, *Dialister*, *Gemella*, and *Granulicatella* were able to distinguish the sample groups. Studies have found different microbial communities in samples collected from the surfaces of oral cancers and matched normal tissues. Samples of 15 oral cancer patients, for example, were enriched with *Firmicutes* and *Actinobacteria* relative to the matched samples, and a single array-based case-control study reported increased counts of specific bacteria in oral cancer (e.g. *Capnocytophaga gingivalis*, *Provatella melaningtoenica*, and *Streptococcus mitis*) [59,63]. Despite these discoveries, few studies have explored the oral microbiome in oral carcinogenesis to determine not only a difference between control and cancer samples but also whether the presence of specific microorganisms is related with malignant progression or prognosis. In this way, larger-scale epidemiologically designed cohorts are needed to verify these associations and, especially, to provide suitably high-powered investigations of host-microbiome-environment and oral cancer outcomes as well as to better depict the molecular basis of the transition from homeostasis and synergy to a dysbiosis relation in oral cancer. Elucidation of these differences in community composition provides a basis for further understanding the role of the microbiome in oral cancer pathogenesis and, perhaps, determining the modulation in the microbiome triggered by oral cancer.

Here, besides showing the modulation in microbiome composition and function in OSCC, we have also indicated the clinical value and diagnostic/prognostic power classification of the differential microbial communities that were identified in the study.

Still, information from metagenomics is limited to the identification of gene, taxa, and inferred functional and structural capacity; thus, additional omics approaches, including metaproteomics, metabolomics, and metatranscriptomics, can increase our knowledge of microbial activity in different disease stages. Researchers need to make efforts, then, in the direction of better exploring these factors that influence the microbiome and identifying the microbiome factors that may influence the host and environmental composition. In this scenario, we have further characterized the metaproteomic profiles of the same saliva samples that were submitted to metagenomic analysis.

Our results show that when considering only the 19 selected OTUs (Fig. 5), we were able to generate a protein database containing both host and microbial protein sequences merged together for proteomic data analysis of cellular and supernatant saliva fractions. The results provide insight into how metaproteomic investigation may result in the determination of factors that contribute to pathology within the oral cavity (Figs. 6–7).

Future studies investigating not only the expression of proteins in saliva but also the glycosylation states may lead us to interesting mechanisms that modulate the oral microbiome and may interfere with the health-dysbiosis balance. In fact, recently, it has been shown that the loss in mice of *Galnt3,* the major *O*-glycosyltransferase expressed in the secretory cells of salivary glands, resulted in changes in the structure, composition, and stability of the oral microbiome [64]. They also identified the major secreted salivary mucin, Muc10, as an in vivo substrate of Galnt3. Given that mucins and their *O*-glycans are known to interact with various microbes, the results suggested that loss of *Galnt3* decreases glycosylation of Muc10, which alters the composition and stability of the oral microbiome. Little is known regarding how individual proteins influence the establishment and composition of the microbiome, and vice versa. Collectin, for example, has been shown to be an abundant protein in the gut microbiota of colorectal cancer tumors and promotes colon tumorigenesis in mouse models. A recent study has also shown a direct link between exposure of human intestinal epithelial cells to colibactin and two unique mutational fingerprints found in human colorectal tumors [65].

Considering that the microbiome may influence the prognosis of OSCC patients and vice versa, knowledge of the microbiome may assist patient risk determination and treatment. It is relevant, then, to promote a more detailed investigation of the salivary microbiome composition in these conditions, evaluate its role in the composition of human proteins, and assess the functional mechanisms related to tumor development. Interestingly, previous research from our group indicated that protein analysis of saliva from OSCC patients resulted in the enrichment of mechanisms associated with the response to the immune system. This can be observed regardless of the presence of the active tumor lesion, suggesting that surgical removal is not sufficient to recover saliva health protein composition. In addition, it is important to point out that patients with surgical removal should also be monitored regarding other treatments, such as radiotherapy, which may lead to complications and induce oral damage (e.g., mucosite and xerostomia) that may have an impact on salivary functionality. Future studies should collect information regarding these procedures and their effects on patients' saliva [66,67].

Although efforts have been made in this study to highlight the cross-talk between the microbiome and oral pathology, this cause-effect relation has yet to be addressed by new studies. Some studies have demonstrated an increased number of microbes in cancer patients [62,68]; in these studies, *Helicobacter pylori* was classified by the World Health Organization (WHO) as carcinogenic to humans and related to gastric cancer [69]. In this way, some interest has arisen regarding the possible relationship between these bacteria and oral cancer. However, this is not a common bacterium in the oral cavity [70]. Regarding the normal commensal microorganisms of the oral cavity, there is a consensus suggesting that the bacteria species are not directly the cause of oral cancer, but it is known that they act synergistically with common risk factors for oral cancer, such as alcohol and tobacco use [62,68].

Furthermore, it is known that tobacco and alcohol may alter the oral microbiome. Alcohol itself is not carcinogenic. However, its metabolic product, acetaldehyde, is highly carcinogenic, despite being classified by the WHO's International Agency for Research on Cancer as a chemical substance whose carcinogenicity to humans was doubted (Group 2B) [71]. Bacteria from oral mucosa, such as *Neisseria*, *Streptococcus*, *Actinomyces*, *and Prevotella*, are involved in the production of acetaldehyde by oxidation of ethanol [71]. In oral epithelial human cells, acetaldehyde can cause sister-chromatid exchanges, gene mutations, and DNA-strand breaks [62,71]. Moreover, chronic smoking alters the abundance of some bacteria (e.g., *Lactobacillus*, *Veillonella*, *Streptococcus*, *Bifdobacterium*, *Atopobium*, *Neisseria,* and *Prevotella* species) [72], leading to an increase in acetaldehyde production from ethanol by the microbiome. Thus, alterations in the oral microbiome associated with oral cancer risk factors contribute to oral cancer development and progression.

In addition, there may be a cross-reactivity between microbial and tumor antigens shaping T-cell repertoires and/or microbial products stimulating pattern recognition receptors that influence the type and intensity of immune responses and, perhaps, tumor differentiation/patterns of invasion. Understanding how the microbiome impacts natural cancer immunosurveillance as well as treatment-induced immune responses will pave the way for more effective therapies and prophylactics [73]. However, the studies to date have primarily demonstrated relationships rather than causative ones [74]. Recently, it has been shown that intratumor bacteria are mostly intracellular and are present in both cancer and immune cells. The authors also noted correlations between intratumor bacteria or their predicted functions with tumor types and subtypes, patients' smoking status, and the response to immunotherapy [75]. In fact, the gut microbiome has been shown to have multiple effects on tumor biology, such as the transformation process, tumor progression, and response to anticancer therapies, including immunotherapy [76]. Thus, characterization of the tumor microbiome may be an essential step in unraveling the effects that tumor bacteria have on different cancer hallmarks. Finally, we have shown that a strategy that combines 16S rDNA sequencing, filtering OTUs by feature attribute, and metaproteomics of the same set of samples may lead to different but complementary information.

5. Conclusions

In summary, the present study has characterized and compared the microbiome of control individuals and OSCC patients with and without active tumors. Moreover, it was capable of showing that both OTUs and proteins, from bacteria and from hosts, are associated with clinical characteristics, highlighting once again the dynamics of microbiome and biological components in different conditions. Future experiments should be directed toward the promising microbial species, such as *Lactobacillus reuteri*, and the characterization of their causal effects in oral cancer phenotypes; these studies should also unravel the underlying mechanisms. In addition, understanding the host's phagocytosis response (neutrophil and macrophage) to the changes in the oral microbiome caused by OSCC combined with the antimicrobial activity of the eleven modulated saliva host proteins may be an important strategy for accessing the role of the microbiome in immunotherapy response as well as promoting the design of novel, targeted antimicrobial drugs for cancer therapy [77]. There is a tremendous opportunity for the microbiome to be a prognostic biomarker, but more specific studies are required.

Data availability

The mass spectrometry proteomic data have been deposited in the ProteomeXchange Consortium (http://proteomecentral.proteomexchange.org) via the PRIDE partner repository [78] with the dataset identifier PXD022859. All other data supporting the findings of this study are available from the corresponding author on request. The metagenomic data was deposited in NCBI repository under the Bioproject submission ID:PRJNA700849 and Biosample submission ID:SUB9044683.

Funding and additional information

Funding was provided by FAPESP Grants (2009/54067-3; 2010/19278-0, 2016/07846-0; and 2018/18496-6 to A.F.P.L.; 2014/06485-9; 2018/15535-0 and 2019/18751-9 to D.C.G.; 2019/21815-9 to A.F.B.L.; 2018/11958-4 to L.X.N.; 2018/02180-0 to C.M.C.; 2018/12194-8 to L.D.T.; 2018/15728-3 to JOS; 15/50590-4 to F.M.S.) and CNPq Grants (470549/2011-4 and 305851/2017-9 to A.F.P.L., 150221/2014-2 to D.C.G.; 305748/2017-3,428527/2018-3, and 306279/2020-7 to F.M.S. FAPESP (15/50590-4) and CNPq306279/2020-7).

Credit author statement

D.C.G. and A.F.P.L. designed research; A.R.S.S., M.A.L., G.C.J., T.B.B., A.C.P.R., collected, processed and stratified clinical samples; L.X.N., D.C.G., S.Y., D.A.A.P. and C.R. performed research; D.C.G., L.X.N., L.D.T., C.M.C., A.F.B.L., B.P., R.R.D., G.P., G.P.T., F.M.S.P., G.T. and F.M.S. analyzed data; D.C.G., L.D.T., C.M.C. and A.F.P.L. wrote the manuscript.

Acnowledgments

We also thank the contribution and support given by the Mass Spectrometry (MAS) Laboratory at Brazilian Biosciences National Laboratory, and NGS Sequencing facility at Brazilian Biorenewables National Laboratory, CNPEM, Campinas, Brazilfor the support on 16S sequencing and processing.

Declaration of Competing Interest

The authors had no conflicts of interest concerning the topic under consideration in this article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbapap.2021.140659.

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